

Weeks 2-10

Grades 2-3

Small counts and concrete arguments; adult reading help is welcome.

Grade bands are entry points. Begin with objects, then try to explain what happens.

Give one page at a time. The packet is a menu, not a checklist for one hour.

Keep facilitator solutions separate from student pages.

Week and investigation	PDF pages
02 Switches and lamps	2-9
03 Repeating secret machines	10-12
04 Around the ring	13-15
05 Tower cities	16-18
06 Code-breaking with guarantees	19-21
07 Take-away games	22-24
08 Rooks and piles	25-28
09 Bouncing paths	29-32
10 Bridges and delivery routes	33-36

Print US Letter, single-sided, at 100% / Actual Size.

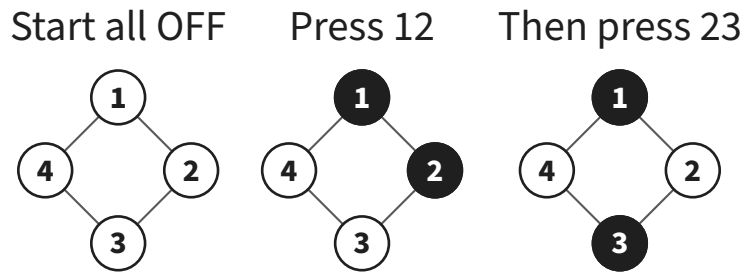
Page ranges above count this cover as page 1. Bookmarks open each week.

Weekly footers retain their own page numbers and activity IDs for the use log.

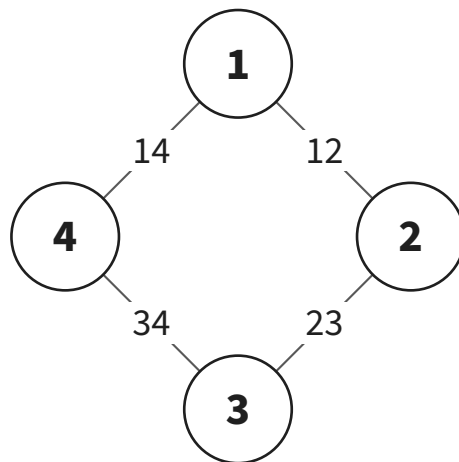
Unpiloted. Week 2 updated September 28; other core revisions September 27.

Problem 1: Put a two-sided counter on each big lamp, white side up: all OFF. Point to a line. Flip the counters at *both* ends. Dark means ON.

The line joining lamps 1 and 2 is called **12**. Replay these two moves on your big board.



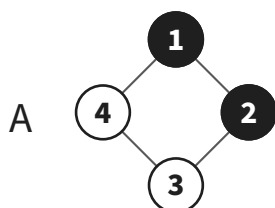
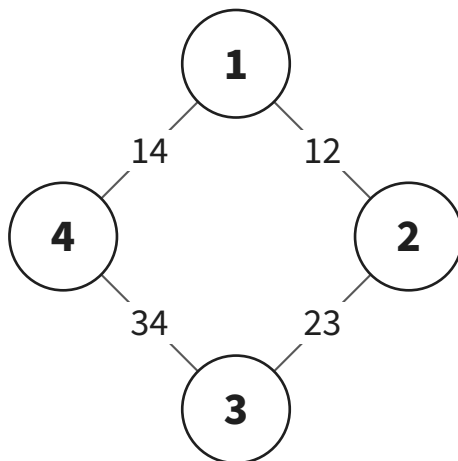
Write the moves in order: **12, 23**.



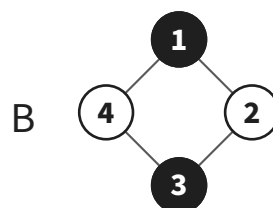
Make the dark lamps white again by flipping pairs. Write your moves so a partner can replay them.

Moves: _____

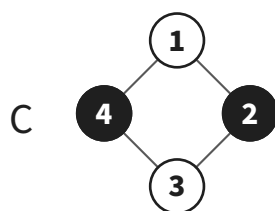
Problem 2: Make each target picture on the big board. Reset all counters to white before each target. Flip both ends of one line at a time. Write your moves in order.



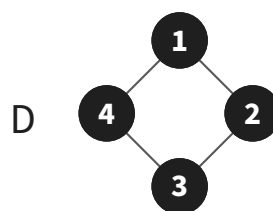
Moves: _____



Moves: _____



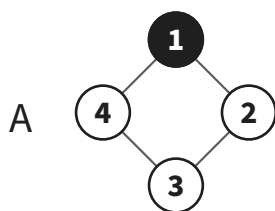
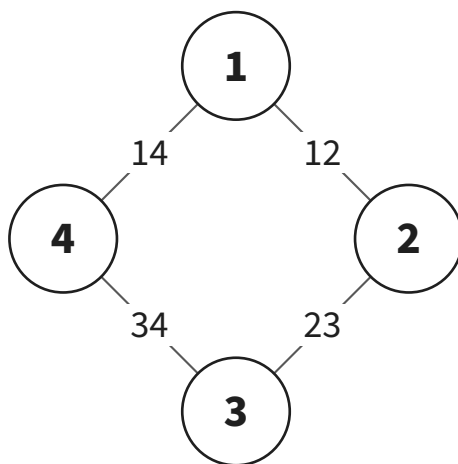
Moves: _____



Moves: _____

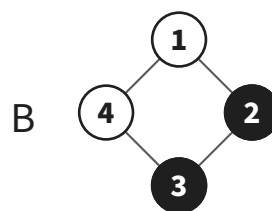
Give a partner one of your move lists. Reset all OFF. Have your partner replay it and point to its target.

Problem 3: Try each target picture. Start all OFF each time. Flip both ends of one line at a time. Write a move list that works, or check “not yet.”



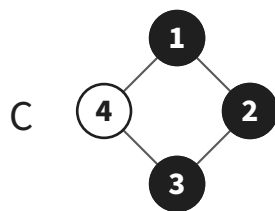
Moves: _____

☐ Not yet



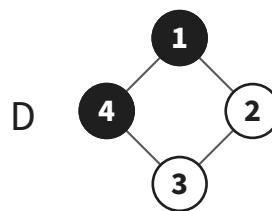
Moves: _____

☐ Not yet



Moves: _____

☐ Not yet

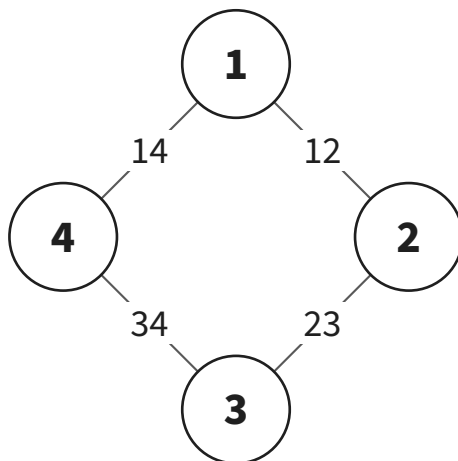


Moves: _____

☐ Not yet

For a “not yet” picture, try a different move list on the board. Keep any list that works.

Problem 4: For each row, turn counters over by hand to set up its starting picture on the big board. Leave lamps 3 and 4 OFF. Then press 12 once. Darken the ON lamps in the answer picture. Count ON lamps before and after.



A  — 12 — 
ON before: _____

press 12 →

 — 12 — 
ON after: _____

B  — 12 — 
ON before: _____

press 12 →

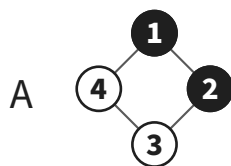
 — 12 — 
ON after: _____

C  — 12 — 
ON before: _____

press 12 →

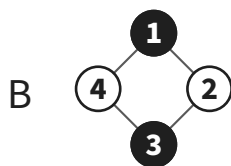
 — 12 — 
ON after: _____

Problem 5: These are the targets you tried. For each picture, circle “made” or “not yet.” Count its ON lamps. Use the big board to try any picture again; reset all OFF each time.



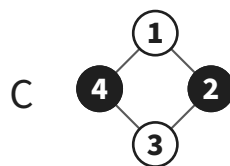
ON: _____

made / not yet



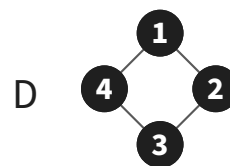
ON: _____

made / not yet



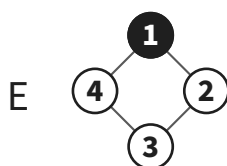
ON: _____

made / not yet



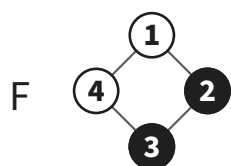
ON: _____

made / not yet



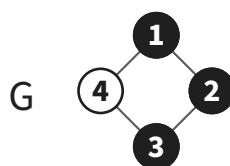
ON: _____

made / not yet



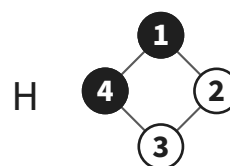
ON: _____

made / not yet



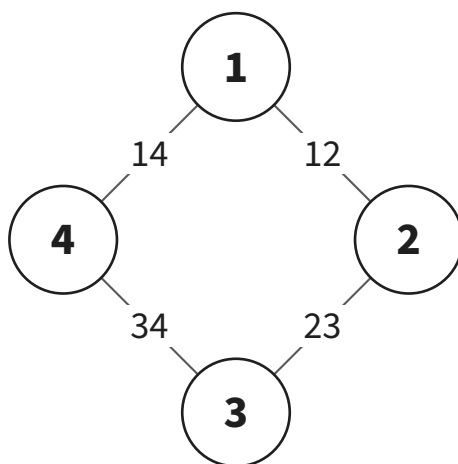
ON: _____

made / not yet



ON: _____

made / not yet

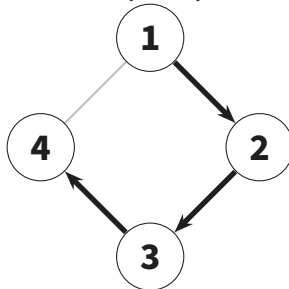


Compare the ON counts in the pictures you made.

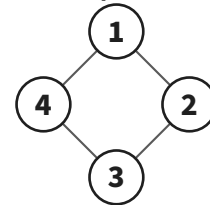
Point to what is alike. Then compare the “not yet” pictures.

Problem 6: Follow each arrow path on the big board: flip both ends of each line, in order. Start all OFF for each path. Darken the ON lamps in its final picture.

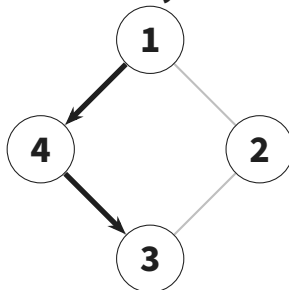
A: 12, 23, 34



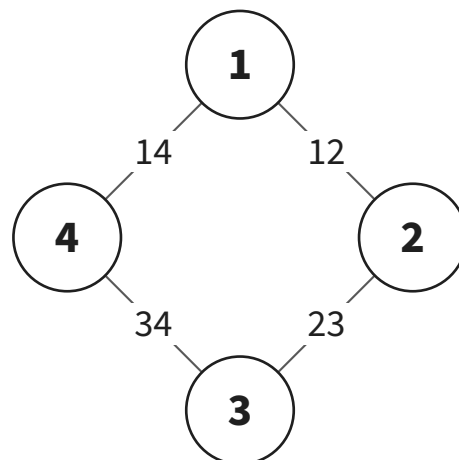
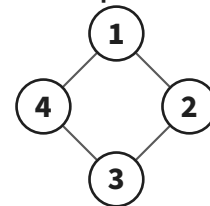
Final picture:



B: 14, 43

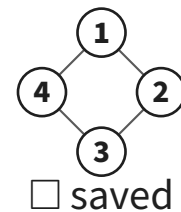
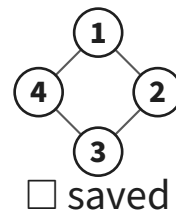
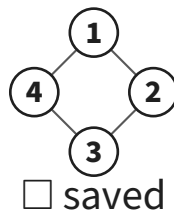
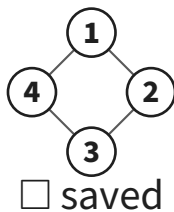
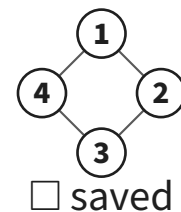
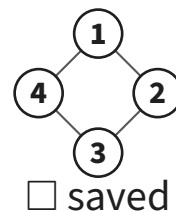
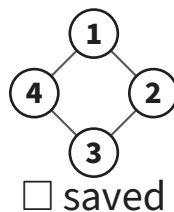
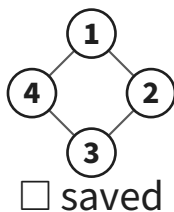
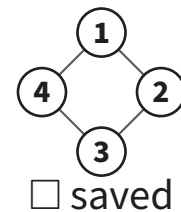
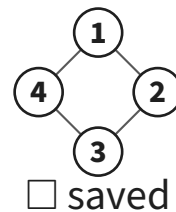
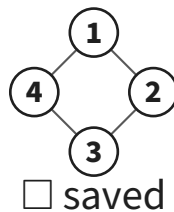
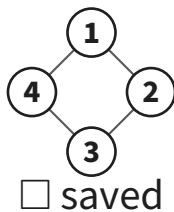
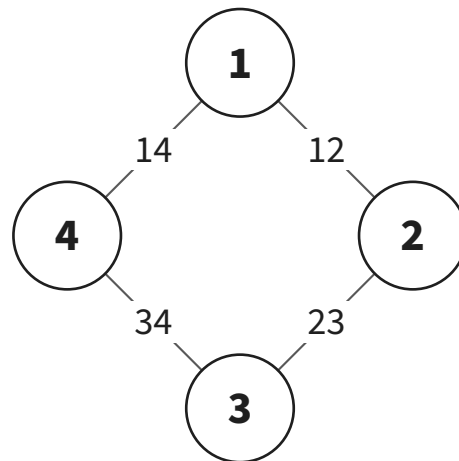


Final picture:



Replay one path slowly. Point to each lamp that gets flipped twice. Point to each lamp flipped just once.

Problem 7: Start all OFF on the big board. Try some moves. Copy the result onto a little ring: darken its ON lamps. Reset and make different pictures. Include all OFF. Leave extra rings empty.



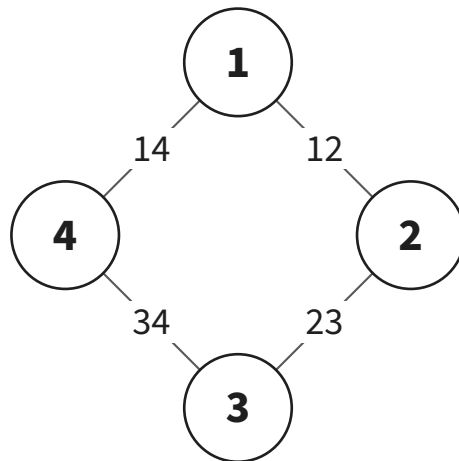
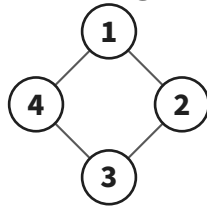
Check “saved” under each picture you record, including all OFF. This keeps unused rings separate.

Problem 8: Look at the pictures you saved in Problem 7. Count the ON lamps in each. Write that count beside its little ring. Point to the pictures with the same count. Count how many different pictures you have.

I saved _____ different pictures.

Choose a picture whose moves you have not written down. Copy it here. Start all OFF on the big board and find a move list that makes it.

My target:



My moves: _____

Reset all OFF and ask a partner to replay your list. Compare the result with your target.

Problem 1: A key changes each letter while keeping its place in a message. Use key P below to encode all three messages. Every output letter appears once in the key, so you can also follow it backward.

Input	A	B	C	D
P output	B	C	A	D

Message	Encoded message
ABBA	
ADAD	
ABCD	

Problem 2: Decode CAAC and DBDB. Check each result by encoding it again.

Problem 3: Invent a key using A,B,C,D once each as outputs. Encode ABBA. Compare it with P’s output. Mark the positions that match each other in both outputs.

Input	A	B	C	D
My output				

Problem 4: Start with ABCD. Change every letter at once using P. Apply P to the new message, and keep going. Record each row. Circle the first turn after 0 when the entire row returns to ABCD. Do the same with Q.

Input	A	B	C	D
P	B	C	A	D
Q	B	A	D	C

Turn	P message	Q message
0	ABCD	ABCD
1		
2		
3		
4		

Problem 5: In each key, follow A until it comes back. Draw arrows between the letters it visits, closing the loop. Choose a letter not yet drawn and repeat. A letter that stays put gets an arrow back to itself. Draw all the loops for P and Q.

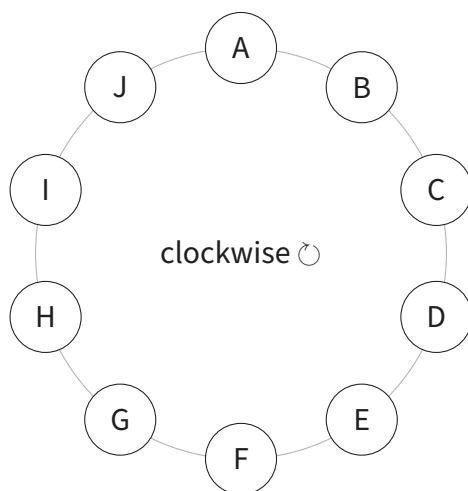
Problem 6: Make a key that takes four turns to return the whole message ABCD. Build it with cards, draw its arrows, then test all four turns.

Problem 7: Use four letter cards to build each arrangement below. An arrow loop must return to its own start. For each arrangement, test the first turn when all four cards are home. Write your key or draw the loops.

Loop sizes	My key or loops	First return
3 + 1		
2 + 2		
2 + 1 + 1		
1 + 1 + 1 + 1		

Problem 8: Compare these four arrangements with your four-turn key. Try to build a six-turn key with four letters. Then add E and try a loop of three letters and a loop of two. Record what happens.

Problem 1: Use a counter on this ring. A turn moves a fixed number of places clockwise; after J comes A. Try a two-place turn from A, then from I. Record the two landings. Count places moved, not the starting place.

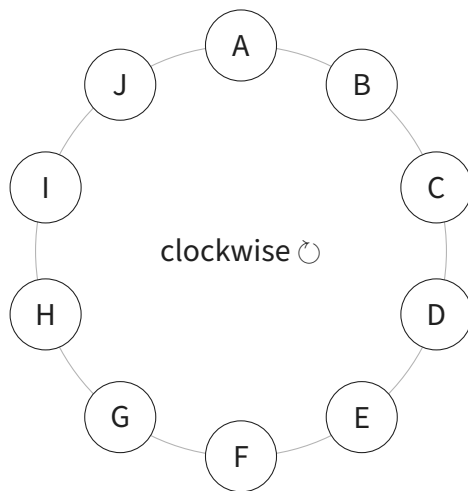


Problem 2: A secret turn takes A to D. Find its step size. Use that same turn from B, I, and J. Record the three outputs.

Problem 3: Encode BAG by shifting each letter with the secret turn. Decode GDG by undoing that turn. Check both messages by reversing your moves.

Problem 4: Someone says B becomes F with that same key. Test the claim on the ring. Record what B actually becomes.

Problem 5: Start at A. Take two clockwise places every turn. Record each landing until A first returns. Repeat from B. Do not record letters you pass over.



Start A: A →

Start B: B →

Problem 6: Try the two-place rule from C, then from D. Compare with your first two routes. Circle which route each new start joins.

Start	Joins A's route	Joins B's route
C	yes / no	yes / no
D	yes / no	yes / no

Problem 7: Mark the A-route letters with a dot and the B-route letters with a small cross on the ring. Follow one more turn from each letter and compare its marking with the next landing.

Problem 8: Use the page-2 ring. From A, take three places per turn and record the complete route until A first returns. Compare its visited letters with the two-place routes.

Problem 9: Test five-place turns from A, then from B. Draw the two routes. Compare their lengths with the two-place and three-place routes.

Problem 10: A two-place route skips letters. Test whether you can still undo a two-place move from A, B, I, and J. Find one clockwise step size that brings every test back. Record each out-and-back trip.

Problem 11: Choose a new step size from 1 to 9. Predict whether it visits all ten letters from A. Test and record its route.

Problem 1: Build each row with towers of heights 1, 2, and 3. Count the towers you can see from each end; a taller tower hides a shorter one behind it. Keep the row still when you change ends.

Order, left to right	Seen from left	Seen from right
1 2 3		
2 1 3		
2 3 1		
3 2 1		

Problem 2: Build a city with heights 1, 2, and 3 once in every row and once in every column. Record the heights in the left grid. Swap two whole rows of towers. Check every row and column again, then record the new city in the right grid.

The rows I swapped:

--

Problem 3: The 3 above the first column means three towers must be visible from that end. Build a city with this clue. Record it on the left. Change the city while keeping the clue and the row and column rules true. Record a different city on the right.

3

3

Problem 4: Copy the first column your clue requires. List the possible orders for the top row. For each top row, try to fill the rest of its city using the grids above.

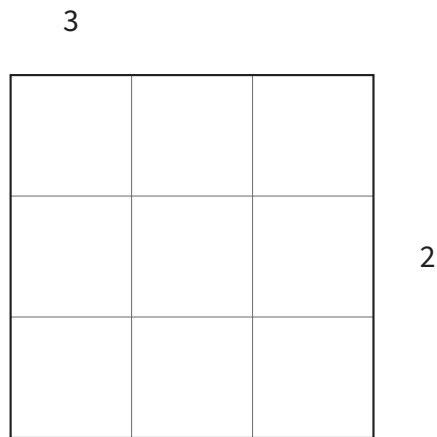
First column, top to bottom:

--

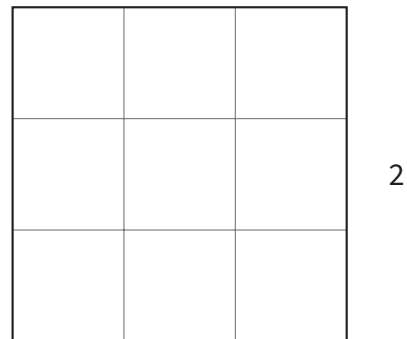
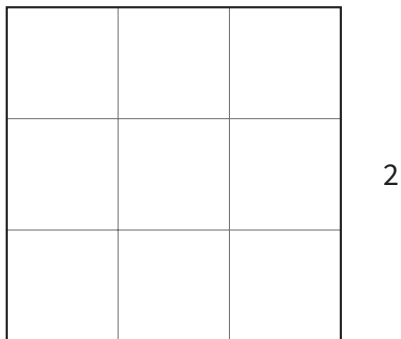
Possible top rows, left to right:

--

Problem 5: Keep the top clue 3. Add a right-hand clue of 2 on the middle row. Test both cities from Problem 3. Cross out any that break the new clue, then record a city that keeps both clues.



Problem 6: Cover only the top clue. Build two different cities that still show two towers from the right of the middle row. Record both and check every row and column.



Problem 1: Use R and B counters; repeats are allowed. A test scores one point for each matching position. Say only the total score. Use three positions. Leave RBR visible and score the tests RRR, BRB, and RBR before hiding any secret.

Test	Score for secret RBR
RRR	
BRB	
RBR	

Problem 2: Hide a three-place secret and play. Keep every test and score visible; add rows on scrap paper if needed. Once only one secret fits, name it; naming costs no test. Reveal the secret and check your record. Swap jobs for another round.

Round 1			Round 2		
Test	Code tested	Score	Test	Code tested	Score
1			1		
2			2		
3			3		
4			4		

Problem 3: Build each visible secret below. Test RRR, BRR, and RBR against it. Compare each changed test with RRR, not with the previous row. Record all three scores.

Secret	RRR score	BRR score	RBR score
BRR			
RBR			
RRB			
BBB			

Problem 4: Hide a new secret. Test RRR, BRR, and RBR and record their scores. Use the score changes to choose R or B for positions 1 and 2. Use the RRR score to find how many Rs still belong in position 3. Name the secret, then reveal and check it.

Test	Code tested	Score
1		
2		
3		

Position 1: Position 2: Position 3: My secret:

Problem 5: Suppose RRR scores 2. Build the three possible secrets BRR, RBR, and RRB. Test each row below against all three. Circle equal scores within a row. Add two tests of your own.

Test	BRR score	RBR score	RRB score
RRR			
BRR			
BBR			
BBB			

Problem 6: Try a different first test, BRB. Make each secret below, then swap R with B at positions 1 and 3 in both the secret and the test. The changed test is RRR. Record the changed secret and check both scores.

Original secret	Score for BRB	Changed secret	Score for RRR
RRB			
BBB			
BRR			

What changed in the positions? What happened to the score?

Problem 1: Two players alternate taking 1, 3, or 4 counters. Taking 2 is not allowed. Never take more than remain. Taking the last counter wins. Play from 5, 6, and 7. Swap who starts and replay each pile. Record the moves in order; a partner may write while you play.

Starting pile	First move	Next moves, in order	Winner: first or second
5			
5			
6			
6			
7			
7			

Problem 2: Build a pile of 2, then a pile of 4. From each pile, try every allowed first move. Reset the pile before the next attempt. Record how many counters each move leaves.

Pile	Allowed move / counters left
2	
4	

Problem 3: Use moves 1, 3, and 4. A position is W if the player about to move has a winning strategy; it is L if every move lets the other player force a win. Put L at 0 because there is no move. Work from small piles upward. List all legal next piles. Mark W if at least one is L; mark L if all are W.

Pile	All legal next piles	W or L	One move to L, if any
0	none	L	none
1			
2			
3			
4			
5			
6			
7			

Problem 4: Build a nonempty pile you marked L, such as 2 or 7. Let your partner choose a legal move, then use your chart to reply. Reset and try every different first move. Record your replies.

Starting pile, each first move, and my replies:

Problem 5: Use moves 1, 3, and 4. From 8 counters, try taking 1, 3, and 4. Reset each time. Use your chart to record what each move leaves and whether that next pile is W or L. Circle every winning first move.

Take	Pile left	Its W/L label
1		
3		
4		

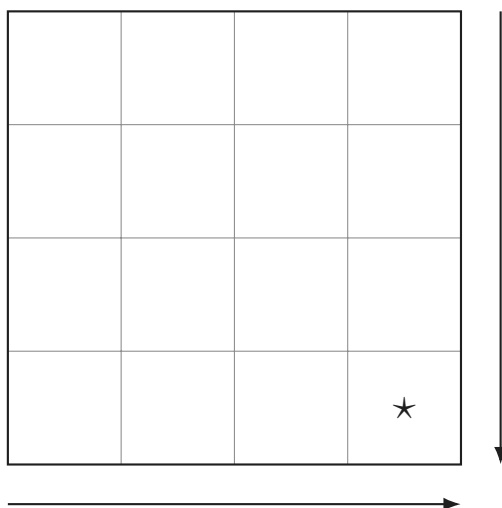
Problem 6: Do the same from 10. First decide the missing labels for 8 and 9, using all their legal moves. Record all winning moves from 10, not just the first one you find.

Take	Pile left	Its W/L label
1		
3		
4		

Problem 7: Choose a starting pile from 8 through 14. Make a strategy card: record each possible first move and the pile it leaves. Circle moves that hand your opponent an L. Test the card in a game.

My pile and strategy card:

Problem 1: Share a token. Take turns moving it right OR down one or more squares. Do not move diagonally. Whoever reaches the star wins. Play the starts below, then swap who starts. A start written (2,1) is 2 squares left and 1 square above the star.



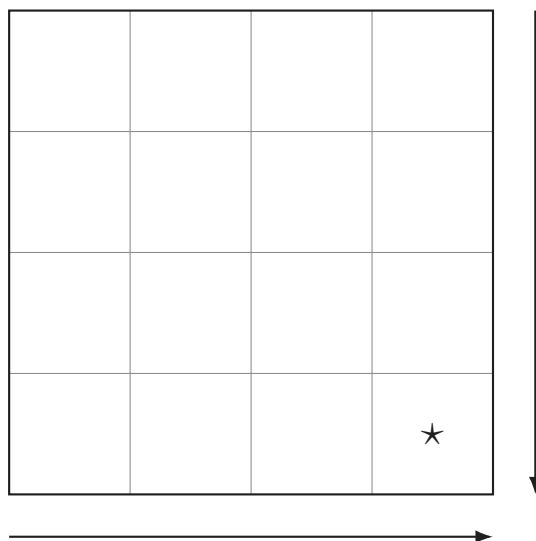
Start	First game: who won?	After swapping: who won?
(1,0)		
(1,1)		
(2,1)		
(2,2)		

Problem 2: Replay one start. Choose a different first move. Record that move and the reply.

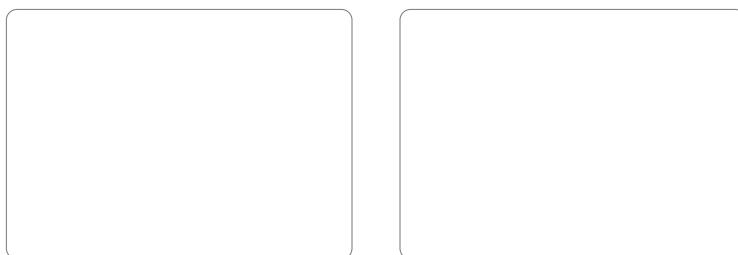
Problem 3: Play from (3,1), (3,2), (2,3), and (3,3) on the board below. For each, write a first move you would choose. Test it against your partner.

Start	My move leaves	Partner's reply leaves
(3,1)		
(3,2)		
(2,3)		
(3,3)		

Problem 4: Make a map of the board. Mark T on a square if its next player cannot win against perfect replies. Mark W if that player can force a win, and ? while unsure. Mark the star T: the game has ended, so its next player has no move. Work outward from the star. A W needs one move to a T; a T has only moves to W squares.



Problem 5: Put the token at (3,2) on your board. Build piles of 3 and 2 in the spaces below. Slide right 1 square and remove 1 from the first pile. Reset. Slide down 2 squares and remove 2 from the second pile. Choose two more slides; make each matching pile move.



My slide from (3,2)

The two piles left

Problem 6: Now play with piles only. On each turn remove one or more counters from one pile. Take the last counter to win. Try each start. If the piles differ, make them equal in one move. If they match, let your partner move and then restore the match. Record a move and reply.

Start	After first move	After reply
(2,2)		
(3,2)		
(1,4)		
(4,4)		

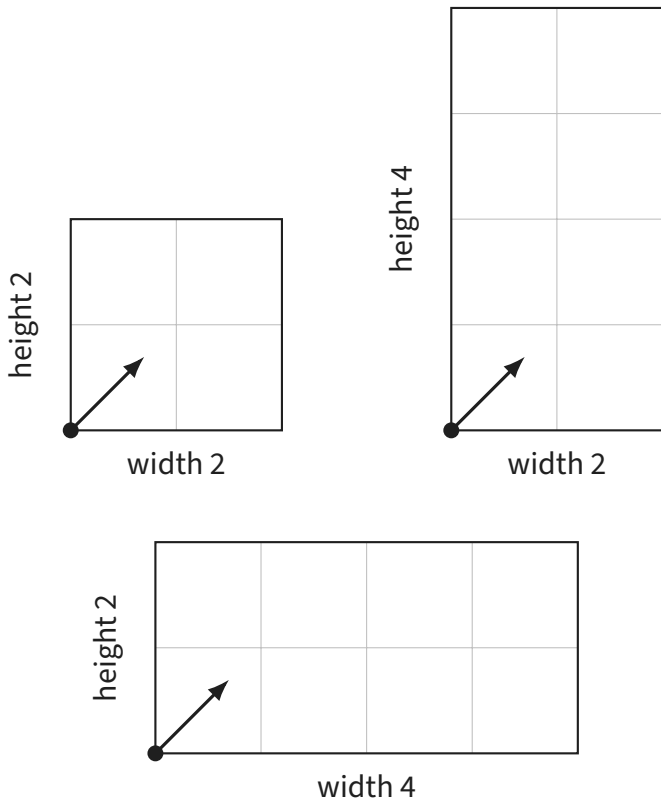
Problem 7: Add a third pile; keep removing from only one pile per turn. Play (1,1,1), then (1,1,2). Draw a first move that leaves two matching piles and one empty pile in each game.

Problem 8: Build (1,2,3). The table gives every possible first move. For each row, rebuild those piles and find a reply leaving two equal piles and one empty pile. Write the three piles after your reply. Keep the pile order unchanged.

After the first move	After my reply
(0,2,3)	
(1,0,3)	
(1,1,3)	
(1,2,0)	
(1,2,1)	
(1,2,2)	

Problem 9: Choose a two-pile start with unequal piles. Ask a partner to leave you equal piles in one move. Record your start and the move.

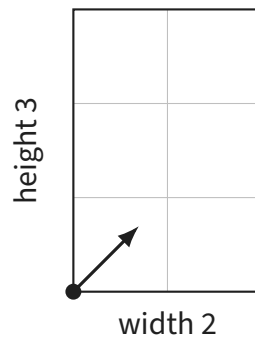
Problem 1: Start at each black dot and trace diagonally up-right, one grid step across for each step up. At a side wall reverse left/right; at a top or bottom wall reverse up/down. Stop at the first corner. Line crossings inside the table do not change the direction. Mark every bounce with a dot and circle the ending corner.



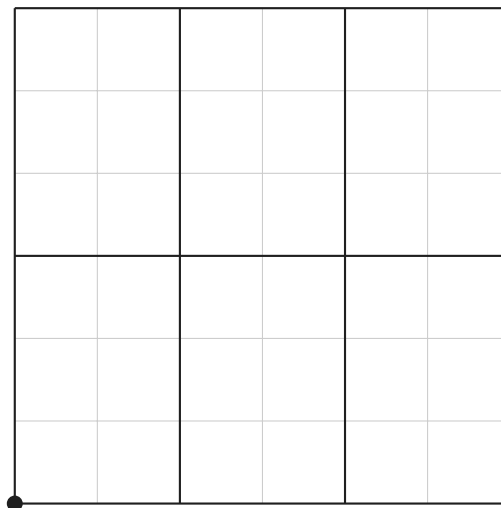
Problem 2: For each table, record the corner and number of bounces. Count wall bounces, not the starting or ending corner.

Width × height	Ending corner	Bounces
2 × 2		
2 × 4		
4 × 2		

Problem 3: Trace this 2×3 table. Mark bounces with dots. Record their order with small numbers beside the dots. Circle the ending corner.



Problem 4: The thick lines below are walls of mirror copies of that table. Instead of bouncing, draw a straight diagonal from the black dot. Stop at the first meeting of a thick vertical and thick horizontal wall. Number the wall crossings before it. Match those numbers to your bounces in Problem 3.



Problem 5: Count the whole room widths and heights from the black dot to your stopping point. Record: widths _____; heights _____; horizontal distance _____.

Problem 6: Draw a straight diagonal to the first meeting of a vertical and horizontal room wall in each picture. The left rooms have width 3, height 6. The right rooms have width 3, height 9. Mark each wall crossing. Count whole widths and heights to the stopping point.

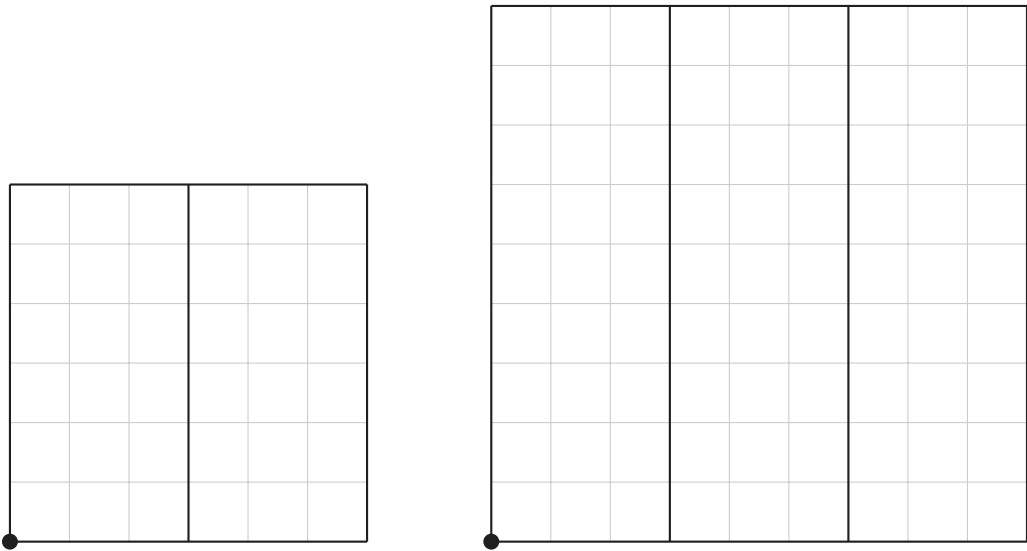


Table	Widths	Heights	Ending corner
3×6			
3×9			

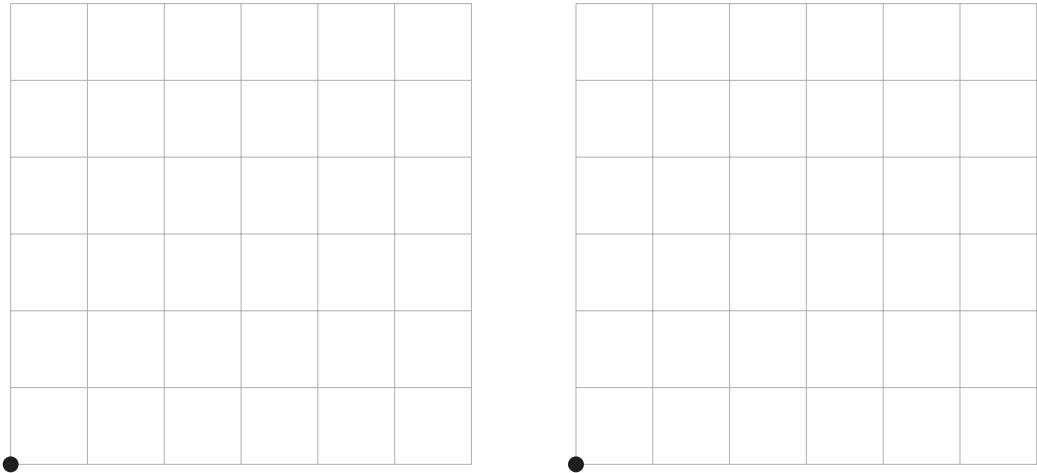
Problem 7: Fold the trip back into its original room. One whole width carries you to the right; the next brings you back left. One whole height carries you to the top; the next brings you back down. Move your finger back and forth along these edges to find the ending side after each count. Use these records to finish the corner column above.

Whole widths or heights	1	2	3	4	5	6
Left/right endpoint	right					
Bottom/top endpoint	top					

Problem 8: Use lists of wall distances to predict a corner. For width 4, list 4,8,12,... For height 6, list 6,12,18,... Stop at their first common number. Do the same for each row. Record whole room counts and use Problem 7 to predict the corner.

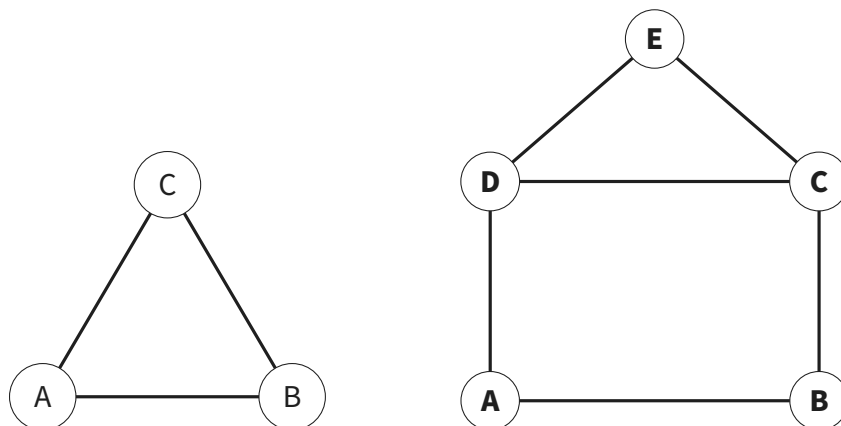
Table	First common distance	Widths	Heights	Corner
4×6				
2×6				
3×3				

Problem 9: On the left grid draw a rectangle that reaches the top-right corner after at least one bounce. On the right draw one that reaches the top-left. Use whole-number widths and heights no larger than 6. Trace each path to test your design.



Problem 10: Record each table's width, height, and ending corner. Swap designs with a partner and test their prediction.

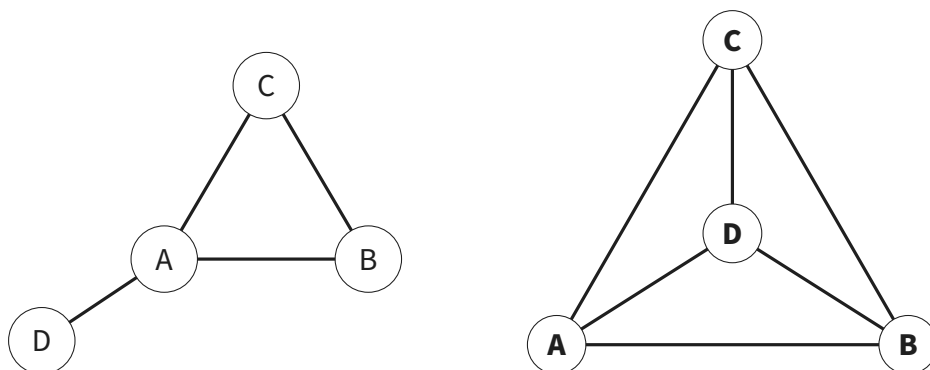
Problem 1: Walk every line once without jumping. You may visit a dot again. Mark each used line with a counter, or number the lines in walking order. Try both maps. You choose the starting dot. Record one route as a list of visited dots, such as A–B–C–A.



Problem 2: Try the house again from A, then from C, then from D. Record a complete route if you find one. Otherwise record the route you tried and the unused lines.

Start	Route	Unused lines
A		
C		
D		

Problem 3: Try to walk each map using every line once. On the right map, D is a junction connected to all three outside dots. Try two different starts on each map. Record where each attempt stops and any unused lines.

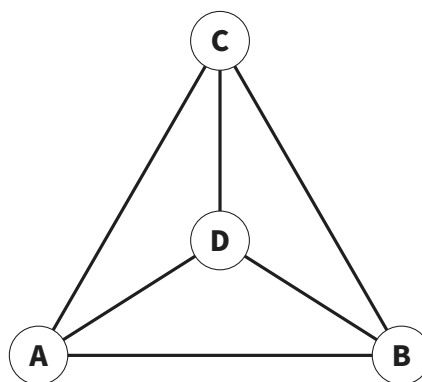
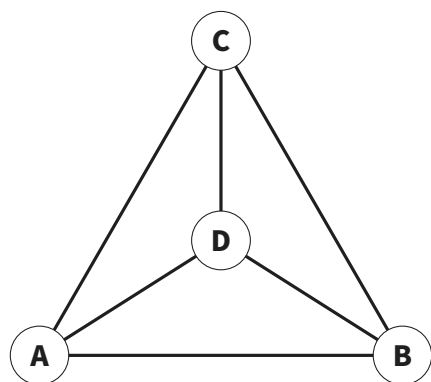


Problem 4: Count the lines meeting at each dot on the house, the triangle with a branch, and the right-hand map above. This count is called degree. Record the counts and circle the odd ones. Compare odd dots with the start and finish of your complete routes.

Map	A	B	C	D	E
House					
Triangle + branch					–
Triangle + center					–

Problem 5: Choose a complete route. At a dot in its middle, pair each arrival with the next departure using two fingers. Record the pairs of lines at that dot.

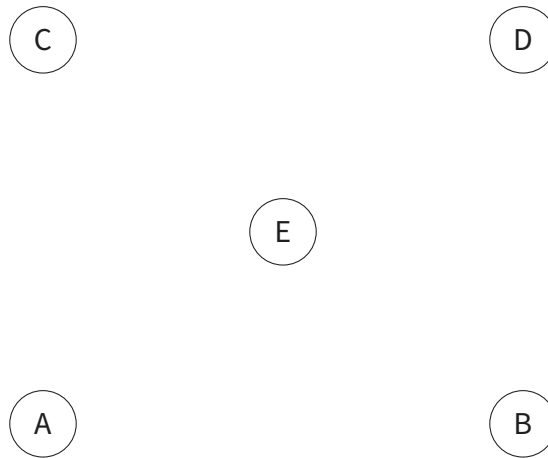
Problem 6: Add a second bridge between A and B on the left map. Draw it curved so the two bridges stay separate. On the right map add a second bridge between A and C. Count the new degrees and circle the odd dots. A crossing without a dot is not a junction.



Problem 7: Find a complete every-bridge-once walk on each repaired map. Start at one of its odd dots. Number the bridges in walking order, keeping both copies separate. Record the two routes.

Problem 8: Choose one repaired map and add a bridge joining its two remaining odd dots. Try to walk every bridge once and return to your start. Record your added bridge and route.

Problem 9: Draw a connected map on the five dots below with exactly two odd dots. Every bridge must join different dots; curved and parallel bridges are allowed. Count each dot's bridges and circle the odd counts.



Problem 10: Find a route using each bridge once. Number its bridges or record its visited dots below. If the attempt gets stuck, try another route or revise a bridge. Keep this solution hidden while a partner tries your map.

Problem 11: Make a second version by adding one bridge. Choose its endpoints so every dot now has even degree. Give your partner the new map and ask for a route that ends where it started. Record the added bridge and a complete route.